UNION

Agenda

- Union
- Defining a Union
- Memory Allocation
- Accessing Union Members
- Structure Vs Union
- References

Union

- Definition – Same as structure
- Syntax – Same as structure
- Difference in structure and union
  - Keyword – union
  - Memory allocation
  - Variable Storage

Union

- A **union** is a special data type available in C that allows to store different data types in the same memory location.
- You can define a union with many members, but only one member can contain a value at any given time.
- Unions provide an efficient way of using the same memory location for multiple-purpose.

Defining a Union

- The union statement defines a new data type with more than one member for your program.
- **Syntax:**

```
union tag_name
 {
     data-type   Field1;
     data-type   Field2;
        ………
 }[one or more union variables];
```

- The tag_name is optional and each member definition is a normal variable definition.
- At the end of the union's definition, before the final semicolon, you can specify one or more union variables but it is optional.

Memory Allocation

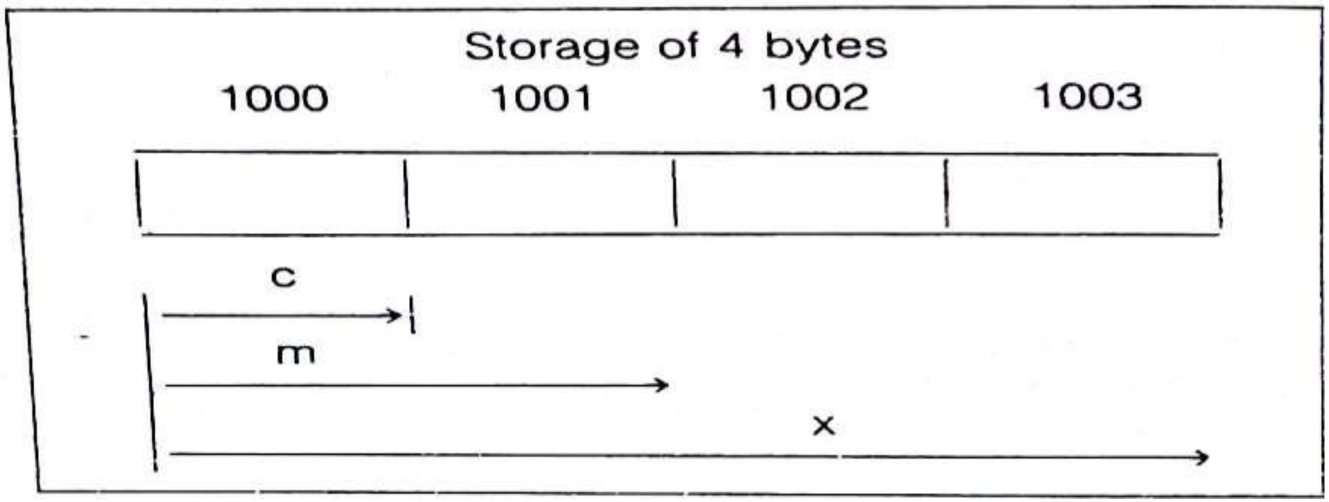
```
union Exdata
{
    int i;
    float f;
    char str[20];
} data;
```

- The memory occupied by a union will be large enough to hold the largest member of the union.
- For example, here, Exdata type will occupy 20 bytes of memory space because this is the maximum space which can be occupied by a character string.

Memory Allocation

```
union Example
{
int m;
float x;
char c;
} u;
```

All members share the same storage area in computers memory.



Example Program1

```
#include <stdio.h>
#include <string.h>

union Exdata
{
   int i;
   float f;
   char str[20];
};

int main( )
{
   union Exdata data;
   printf( "Memory size occupied by data : %d\n", sizeof(data));
   return 0;
}
```

OUTPUT:

Memory size occupied by data : 20

Accessing Union Members

- To access any member of a union, we use the **member access operator (.)**.
- The member access operator is coded as a period between the union variable name and the union member that we wish to access.
- You would use the keyword **union** to define variables of union type.

Example Program 2

```
#include <stdio.h>
#include <string.h>

union Exdata
{
    int i;
    float f;
    char str[20];
};

int main( )
{
    union Exdata data;
    data.i = 10;
    data.f = 220.5;
    strcpy( data.str, "C Programming");
    printf( "data.i : %d\n", data.i);
    printf( "data.f : %f\n", data.f);
    printf( "data.str : %s\n", data.str);
    return 0;
}
```

OUTPUT:

```
data.i : 1917853763
data.f : 4122360580327794860452759994368.000000
data.str : C Programming
```

Note:

Here, we can see that the values of **i** and **f** members of union got corrupted because the final value assigned to the variable has occupied the memory location and this is the reason that the value of **str** member is getting printed very well.

Example Program 3

```c
#include <stdio.h>
#include <string.h>

union Exdata
{
    int i;
    float f;
    char str[20];
};

int main( )
{
    union Exdata data;
    data.i = 10;
    printf( "data.i : %d\n", data.i);
    data.f = 220.5;
    printf( "data.f : %f\n", data.f);
    strcpy( data.str, "C Programming");
    printf( "data.str : %s\n", data.str);
    return 0;
}
```

OUTPUT:

```
data.i : 10
data.f : 220.5
data.str : C Programming
```

Note:
Here, all the members are getting printed very well because one member is being used at a time.

Structure Vs Union

Similarities

1. Both are user-defined data types used to store data of different types as a single unit.
2. Their members can be objects of any type, including other structures and unions or arrays. A member can also consist of a bit field.
3. Both structures and unions support only assignment = and sizeof operators. The two structures or unions in the assignment must have the same members and member types.
4. A structure or a union can be passed by value to functions and returned by value by functions. The argument must have the same type as the function parameter. A structure or union is passed by value just like a scalar variable as a corresponding parameter.
5. **'.'** operator is used for accessing members.

Differences

	STRUCTURE	UNION
Keyword	The keyword **struct** is used to define a structure	The keyword **union** is used to define a union.
Size	When a variable is associated with a structure, the compiler allocates the memory for each member. The size of structure is **greater than or equal to the sum of sizes of its members.**	when a variable is associated with a union, the compiler allocates the memory by considering the size of the largest memory. So, size of **union is equal to the size of largest member.**
Memory	Each member within a structure is assigned unique storage area of location.	Memory allocated is shared by individual members of union.
Value Altering	Altering the value of a member will not affect other members of the structure.	Altering the value of any of the member will alter other member values.
Accessing members	Individual member can be accessed at a time.	Only one member can be accessed at a time.
Initialization of Members	Several members of a structure can initialize at once.	Only the first member of a union can be initialized.

Excercise

- Write a C Program to display employee details using union.
- Tip:

```
union employee
{
        int code;
        char name[10];
        int bpay;
}u[5];
```

References

- https://www.tutorialspoint.com/cprogramming/c_unions.htm#:~:text=A%20union%20is%20a%20special,memory%20location%20for%20multiple%2Dpurpose.

THANK YOU